

NeuroShield: A Low-Cost, Personalized, and Explainable Wearable for Early Detection of Physiological Stress

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Abstract

Continuous monitoring of physiological arousal is now practical with consumer wearables, yet most deployed systems report instantaneous summaries relative to population norms, expose little of their internal reasoning, and rely on research-grade sensing whose behavior on inexpensive hardware is poorly characterized. We present *NeuroShield*, a low-cost wearable system for the early detection of acute stress and cognitive load that integrates electrodermal activity (EDA), photoplethysmography (PPG), skin temperature, and inertial sensing with per-user baseline normalization and per-signal explanation. We describe the end-to-end architecture—sensing front end, on-device compute, and companion application—and a machine-learning methodology in which a family of compact, interpretable models is matched to established public corpora (WESAD, SWELL-KW, PPG-DaLiA, the Nurse Stress dataset, Stress-Predict, and a longitudinal consumer activity dataset). We specify an evaluation protocol that quantifies the accuracy retained when transferring from research-grade to low-cost hardware and the marginal contribution of per-user personalization, both under leave-one-subject-out validation, and we describe a primary data-collection protocol on the target hardware. Because no public corpus contains labelled panic, heat-exhaustion, or burnout episodes, all models are trained against proxy labels, and the system is positioned as a reproducible research prototype rather than a clinical instrument. This paper contributes the system design, the model–dataset matching, and the evaluation methodology; empirical results are the subject of ongoing work.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; • **Computing methodologies** → *Machine learning*.

Keywords

wearable sensing, stress detection, electrodermal activity, photoplethysmography, personalization, explainable machine learning, low-cost hardware

1 Introduction

Acute stress and cognitive overload produce measurable changes in the autonomic nervous system, including elevated electrodermal activity (EDA), shifts in heart rate and its variability, and changes in peripheral skin temperature. Consumer wearables have made the acquisition of such signals widespread, but for the task of *early* warning they exhibit three recurring limitations. First, they report the current state rather than flag an escalation as it develops. Second, they compare measurements against population norms

despite substantial inter-individual variation in baseline physiology. Third, they expose a single opaque score with no account of which signal drove a given output. A fourth, more fundamental question concerns hardware: the modalities most diagnostic of arousal, particularly finger and palmar EDA, are best characterized on research-grade instruments, and the degree to which low-cost, open hardware preserves detection quality is not well established [9, 10].

This paper addresses these gaps with *NeuroShield*, a wearable system whose design couples inexpensive, off-the-shelf sensing with two principles typically absent from consumer devices: decisions are normalized against each wearer’s own resting baseline, and every alert is accompanied by an attribution to the physiological signals responsible for it. The contributions are as follows.

- An integrated, low-cost wearable architecture combining four complementary physiological modalities with on-device per-user normalization and per-signal explanation.
- A model family in which each compact, interpretable classifier is matched to an appropriate public corpus, with a corresponding feature representation and personalization scheme.
- An evaluation protocol that quantifies cross-hardware transfer and the marginal benefit of personalization under leave-one-subject-out validation, plus a protocol for primary data collection on the target hardware.
- A delineation of the scope and limitations of physiological stress inference under proxy labels, positioning the system as a reproducible research prototype on open hardware.

2 Related Work

NeuroShield draws on four lines of research: physiological stress sensing, machine learning for affect detection, model personalization, and explainable machine learning.

Physiological sensing of stress. Sympathetic activation elicits changes in EDA, heart rate variability (HRV), and skin temperature. EDA, reflecting eccrine sweat-gland activity, is among the most sensitive indices of arousal and is conventionally recorded at the fingers or palms [9]. Recording site materially affects fidelity: van Dooren et al. compared sixteen locations and found the fingers among the most responsive [10]. Electrode material admits low-cost alternatives: Haddad et al. demonstrated dry silver/silver-chloride e-textile electrodes tracking EDA with high fidelity at the finger [12]. Arousal also manifests thermally: infrared measurements show that stressor intensity predicts the extent of peripheral skin cooling through vasoconstriction [13]. These findings motivate *NeuroShield*’s finger-based placement and its use of skin temperature as a corroborating modality.

Datasets and benchmarks. Public corpora have driven progress in wearable affect recognition. WESAD [1] pairs a chest device and a wrist-worn Empatica E4 across fifteen subjects under the Trier Social Stress Test. SWELL-KW [3] targets office cognitive load; PPG-DaLiA [2] provides motion-rich PPG with ECG ground truth; the Nurse Stress dataset [4] captures stress during hospital shifts; and Stress-Predict [5] contributes a thirty-five-subject multi-stressor study.

Generalization and the laboratory-to-field gap. Surveys observe that laboratory stress detection is well studied but naturalistic performance remains limited [6]. Prajod et al. report that stressor *type* dominates cross-dataset generalization, whereas measurement-device brand matters comparatively little [11]—directly relevant to low-cost deployments.

Personalization. Subject-independent models often underperform because baseline physiology varies. Taylor et al. showed that personalized multitask learning substantially improves mood and stress prediction [7]. In a lighter-weight approach, Lamichhane et al. introduced a per-subject normalization factor from a physiological reference point, reducing error in free-living monitoring [15]. NeuroShield adopts a transparent variant through per-user baseline normalization.

Explainable machine learning. Most consumer devices emit a single closed score. Additive attribution methods such as SHAP [8] attribute a prediction to its inputs, and recent work applies such techniques to wearable stress classifiers, finding the relied-upon features align with known electrodermal responses [16]. NeuroShield surfaces the signals responsible for each alert.

3 System Architecture

NeuroShield comprises four primary sensing front ends (and one optional), an on-device microcontroller that performs feature extraction, per-user normalization, and inference, a communication link, and a companion application (Figure 1). In the self-contained configuration inference executes on the device and only a compact alert payload is transmitted; during development the sensors stream to a host computer over USB or Wi-Fi.

3.1 Sensing and hardware

The system acquires PPG from a MAX30102 over I²C. EDA has no single-chip solution: a small voltage is applied across two finger electrodes and the resulting current is digitized by an ADS1115 sixteen-bit ADC. Skin temperature is read by an MLX90614 infrared sensor, and a BNO055 inertial measurement unit supports motion-artifact rejection. A single-lead AD8232 electrocardiography front end is available as an optional modality for improved heart-rate-derived features; single-lead AD8232 acquisition correlates strongly with standard electrocardiography for autonomic assessment [17], albeit with greater motion sensitivity. Table 1 summarizes the components. Development uses a Saeed XIAO ESP32-S3 (Wi-Fi streaming to a host); the final wearable targets a Saeed XIAO nRF52840 with lower idle power and native Bluetooth Low Energy. Both support on-device inference via TensorFlow Lite Micro or Edge Impulse. Power is a lithium-polymer cell with a charge controller and a 3.3 V regulator; recording-site evidence [10] motivates

Table 1: Core hardware components and their roles.

Component	Role	USD
MAX30102	PPG: HR, pulse variability, SpO ₂	6–12
ADS1115	16-bit ADC digitizing finger EDA	10–15
Ag/AgCl electrodes	Finger EDA contact	7
MLX90614	Skin temperature (IR)	10–16
BNO055	Inertial / motion-artifact rejection	20–35
AD8232 (opt.)	Single-lead ECG (HRV reference)	20–22
XIAO ESP32-S3	Development MCU (Wi-Fi)	7.49
XIAO nRF52840	Final MCU (BLE, low power)	15.90
LiPo + charger + LDO	Power subsystem	8–14

finger-based EDA electrodes, with remaining electronics on the back of the hand.

4 Methodology

4.1 Signal processing pipeline

The pipeline has five stages (Figure 2): acquisition, artifact-aware cleaning, windowed feature extraction, per-user normalization and inference, and explanation. Decisions are expressed on a three-level scale (**Green/Amber/Red**) accompanied by an attribution to the contributing signals.

4.2 Feature extraction and personalization

Each analysis window (5–30 s live; longer overlapping windows for training) is summarized by a feature vector: mean heart rate, short-window pulse-variability statistics, tonic and phasic EDA components, the count of EDA responses, mean skin temperature and its slope, and a motion magnitude from the inertial sensor. Pulse-derived measures are reported as short-window pulse variability rather than clinical HRV, reflecting the limits of optical sensing. To address inter-individual variation, a brief resting recording defines each wearer’s baseline mean and dispersion, and every feature is expressed as a z-score $z = (x - \mu_{\text{base}}) / \sigma_{\text{base}}$. This transparent, inexpensive standardization is a computationally cheap analogue of personalized modeling [7, 15].

4.3 Model family and training data

Rather than a single monolithic model, NeuroShield employs a family of focused, interpretable components, each matched to an appropriate corpus (Table 2). Model M1, the principal classifier, is trained on the wrist (E4) channels of WESAD so that training signals correspond to those acquired by the low-cost front end, and is validated without retraining on Stress-Predict and the Nurse dataset. M2 addresses cognitive load using the HRV and EDA channels of SWELL-KW. M3 recovers heart rate under motion using accelerometer gating—following established accelerometer-referenced artifact-reduction methods that lower heart-rate error under movement [14]—validated against the ECG ground truth of PPG-DaLiA, and governs suppression of inference in high-motion intervals. M4 estimates a thermal-strain trend, and M5 computes longitudinal recovery trends; both are offline analyses. Per-user normalization applies to M1 and M2.

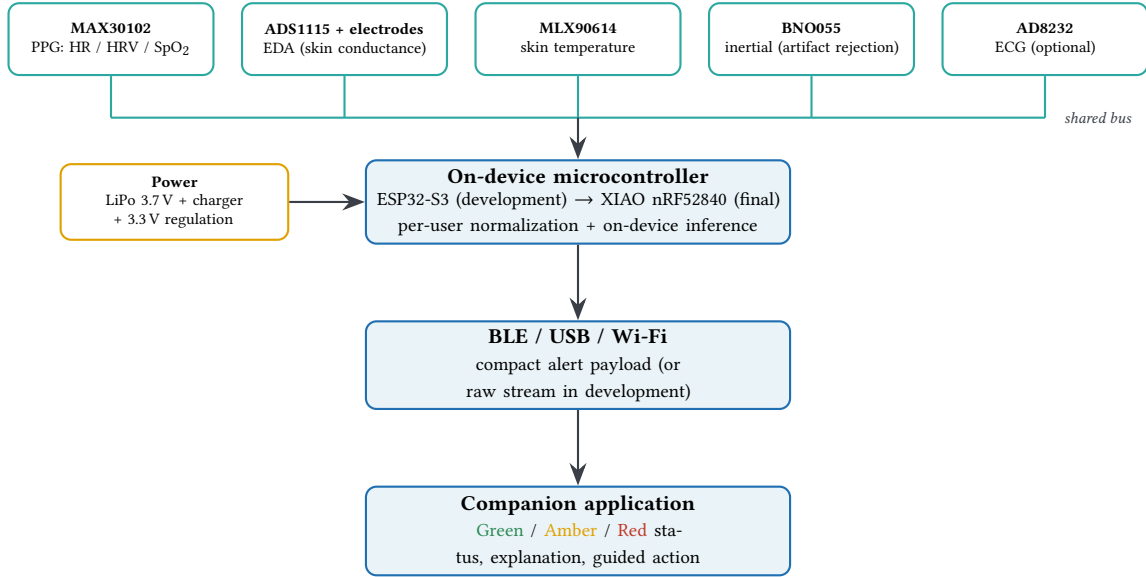


Figure 1: NeuroShield system architecture. Four (plus one optional) sensing front ends share a serial bus to an on-device microcontroller that performs per-user normalization and inference; a compact alert payload is transmitted to a companion application.

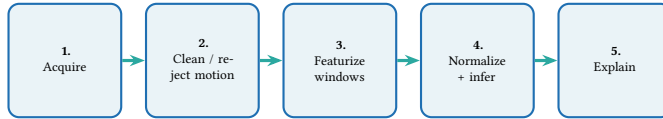


Figure 2: The five-stage signal-processing and inference pipeline.

Table 2: Model family, training/test data, and algorithm.

Model	Task / data	Algorithm
M1	Rest vs. stress; WESAD (test: Stress-Predict, Nurse)	RF / LogReg
M2	Cognitive load; SWELL-KW, WESAD	RF
M3	Motion-robust HR; PPG-DaLiA	Accel. gating
M4	Thermal-strain trend; WESAD (temp, HR)	Trend
M5	Recovery trends; consumer activity data	Trend

4.4 Model selection rationale

Given small sample sizes (fifteen to thirty-five subjects), the requirement for explanation, and on-device execution, ensemble and linear classifiers are preferred over deep architectures, which would be prone to overfitting and would not readily support attribution. Feature attribution (SHAP-style) provides per-signal explanation [8, 16], and leave-one-subject-out evaluation with balanced accuracy and macro-F1 accounts for class imbalance.

Table 3: Public corpora and their roles.

Dataset	Subj.	Role
WESAD	15	Train M1; temp+HR for M4
Stress-Predict	35	Independent test of M1
Nurse Stress	15	Naturalistic test of M1
PPG-DaLiA	15	Train/test M3 (ECG ground truth)
SWELL-KW	25	Train M2 (cognitive load)
Consumer activity	30	Concept data for M5

5 Datasets and Evaluation Protocol

5.1 Datasets

The study uses six public corpora (Table 3). A common constraint governs their use: none contains labelled panic, heat-exhaustion, or burnout episodes; all models are therefore trained and evaluated against proxy labels (acute laboratory stress and daily activity).

5.2 Primary data collection

To evaluate on the target hardware, a small labelled corpus is collected under informed consent from 5–10 participants. Each session follows an acute-stress induction protocol—rest, a Stroop colour-word task, timed mental arithmetic under an auditory distractor, and recovery—with each phase labelled. This corpus is the test set for the cross-hardware experiment.

5.3 Experiments

We specify three experiments (Table 4). E1 quantifies cross-hardware transfer by training M1 on research-grade WESAD signals and

Table 4: Evaluation experiments.

Exp.	Question and procedure
E1	Accuracy retained research-grade → low-cost: train M1 on WESAD, test on locally collected data.
E2	Contribution of personalization: M1 with vs. without per-user normalization (Nurse for naturalistic).
E3	Whether lightweight domain adaptation narrows the gap (stretch).

Table 5: Results reporting template (to be completed; LOSO).

Setting	Bal. Acc.	Macro-F1
M1 on WESAD (LOSO)	TBD	TBD
M1 → Stress-Predict	TBD	TBD
M1 → Nurse	TBD	TBD
M1 → local (E1)	TBD	TBD
E2: with personalization	TBD	TBD
E2: without personalization	TBD	TBD

testing on data collected with the low-cost front end. E2 isolates personalization by comparing M1 with and without per-user normalization, with attention to the naturalistic Nurse data. E3 (stretch) examines whether lightweight domain adaptation narrows the cross-hardware gap. All experiments use leave-one-subject-out validation and report balanced accuracy and macro-F1 per subject with confusion matrices.

5.4 Anticipated results

Empirical results are the subject of ongoing work; this paper contributes the design and protocol. Table 5 is the reporting template to be completed. We expect (i) a measurable but bounded accuracy drop from research-grade to low-cost hardware (E1), and (ii) a positive contribution of per-user normalization that is largest on naturalistic data (E2), consistent with prior personalization findings [7, 15].

6 Discussion, Limitations, and Ethics

Several limitations bound the work. First, construct validity is constrained by label availability: absent labelled panic, heat-exhaustion, or burnout data, all models rely on proxy labels, and the system must not be read as a clinical or predictive medical device. Second, the analog EDA front end (current sensed across finger electrodes via the ADS1115) is the most failure-prone subsystem and requires validation against a reference. Relatedly, optical PPG is known to exhibit reduced accuracy on darker skin tones, a fairness consideration any deployment must acknowledge. Third, on-device deployment imposes model-size and latency constraints, necessitating quantization. Fourth, fabrication of the full wearable requires soldering and flexible-PCB manufacturing with non-trivial lead times. Finally, primary data collection involves human participants and is conducted under informed consent with appropriate handling of physiological data; communication to end users employs

non-alarming language consistent with the system’s status as a research prototype.

7 Conclusion

We presented NeuroShield, a low-cost wearable for the early detection of acute stress and cognitive load that integrates four physiological modalities with per-user normalization and per-signal explanation. The contribution lies less in any single component than in their integration and in an evaluation protocol that quantifies accuracy retained on low-cost hardware and the benefit of personalization, complemented by primary data collection on the target device. By emphasizing transparency, personalization, and reproducibility on accessible open hardware, and by delimiting its claims under proxy supervision, the system is positioned as a reproducible research prototype and a basis for subsequent empirical study.

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